

Pre-feasibility study for the establishment of foundry for molding and metal manufacturing



2017



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1. Executive summary

The project aims to establish a factory of metals melting, casting and forming into different shapes and sizes, to be marketed to the local markets and export markets. Where the factory will be build in AlMafraq governorate and creat new jobs for local country and replacing the imports. The product can be used for different industries such as spare parts of loaders, sewer covers, hummer pins for crushers and another uses.

Based on the propused market share, and the prices considered, the expected revenues of the project estimated at 2,374,800 JD during the first year of the project life and its gradually rise to 2,597,284 JD in the 10th year, the payback period is 2 years and the internal rate of return is 38.92%

Table1 Project Main Indicators

Indicator	Value
Total Investment Cost (Jordan Dinars)	1.419.050
Internal Rate of Return (IRR)	38.92%
Pay-Back Period (Years)	2
Net Present Value (Jordan Dinars)	2.628.130
Job Opportunities provided by the Project	13
Total Revenues (For 10 Years)	24.845.662
Total Operating Expenses (For 10 years)	18.037.951
Total Net Profit (For 10 years)	3.523.698

The most important mineral in the governorate is copper, the estimated reserve of pure mineral is about one metric million ton, the manganese ore reserves estimated about half million metric ton.

Source: Investment Commission, 2015

The most important opportunities Tafeilh governorate:

- Exploiting Uranium to produce atomic energy.
- Industries of building materials
- Chemical industries and fertilizer industry
- Exploiting the oil shale to generate electric energy

Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

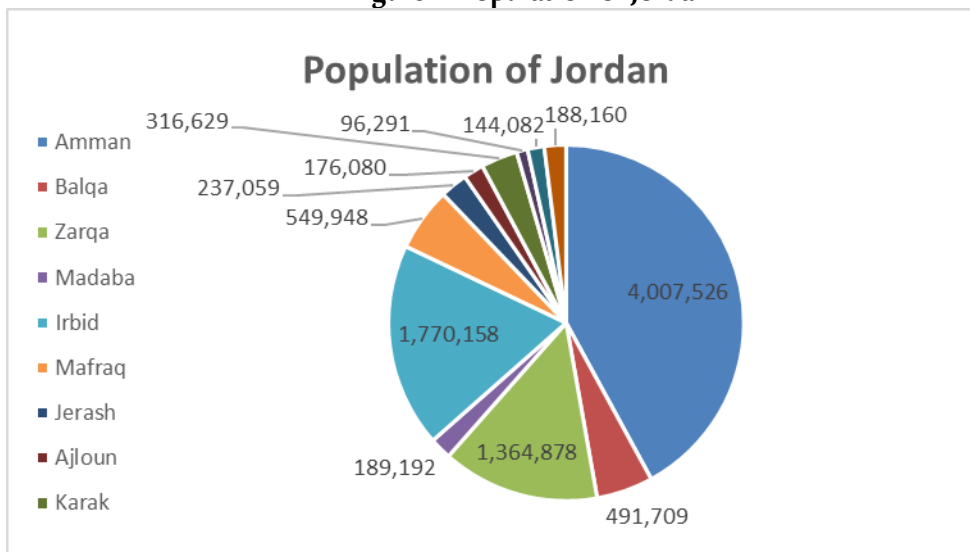
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafrq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table2 Population Distribution according to Age Group:

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

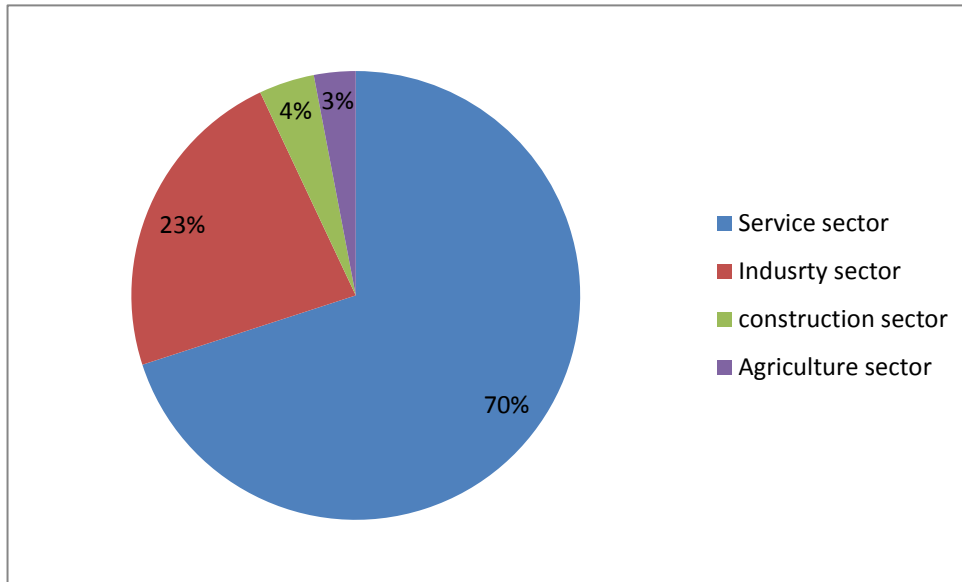
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure2 Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and

medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

furthermore, in its endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

2. Market Analysis

Project Description

The casting and forming metal required many operations such as furnaces, casting moulds based on the required product, in addition to the safety rules. The factory will be established in Mafraq governorate which considered industrial region with a perfect environment for this types of works that will be provided for the local and regional market of factory products. The location of the area can be considered as serves the economic environment of investors and decreasing the transportation cost.

The casting is an industrial process cast the melted materials in moulds equipped with openings and voids according to the required shapes, then the material solidification, the solid part called the casted and it formed the alloys, as: Ipoxxy, concrete, plaster and clay.

The molding process is divided into two categories:

- Molding in consuming moulds: is including sand moulds, crust, plastic, plaster and micro molding, this method is disposalble.

The sand molding: is one of the most important and simple method that used in the past, allow the sandy moulds obtain on casted compare with the permanent moulds with low cost. Are characterized by small scale. The sand also allow the most metals to mold by choice one type of used sand for several day as a production interval to obtain on high production rate (1-20 piece/min for each mould). It is a positive method to produce large pieces. The sand cohesive between it parts by collector material or engines oil also the sand will use for many times and add some material to enhancing the collecting of materials.

- Molding in permanent moulds: this method is use for nonferric metals, it take time for preparing that ranged for many weeks to prepare the steel mould with productivity (5-50 piece/hour for each mould). The weight of the casted materials is ranged between (0.1-9) kg, the viod of the steel mould is coverd by Stalin before the mold process for easy removing the casted from the mould. The permanant moulds have expiry date before damaged or exchanged. In this method an extra in tension strength about 20% and 30% expanding comparing with sand mold, these moulds should be regulary painted. This method is used in (Fe, Al, Mg, Cu).

The metal

Is a compound characterized by a solid phase, in nature it is formed by geological process and it called according to the chemical structure. The properties of metals are varying with each other in structure between the pure metal and simple salts in addition to complicated silicate. The metal is described as a solids material and characterized by its homogeneity between its particles also it has high atomic arrangement with inorganic material, physically it is not easy to decompose it into simple chemical components.

The natural properties of the metals:

- The metals have a special properties that characterized it from another chemical components such as macroscopic:
- The colour: with fixed or variable colour
- The scratch: the colour when the metal is powdered
- Transparency: it ranged between (transparent, semi transparent, dark)
- The luster: the ability of the metal to reflect the light rays, its divided into two parts metallic or nonmetallic.
- Cohesive property: it is the properties that describes the interaction force and divided into:
- The hardness: its a property that resist the scratches.
- The cracking
- The fracture: it's the shape that taken while the external force is applied
- The ability to forming
- The specific gravity
- And other properties such as atomic radiation, electrical, magnetic and melting temperature.

The chemical classifications of the metals:

- Elemental metals such as diamond and graphite
- Sulphides such as galena
- Oxides such as quartz
- Halides such as fluorite
- Phosphate such as apatite

- Carbonates such as calcite
- Sulfates and silicates; quartz

The melting point

Known as a temperature degree that the material is start converting from solid to liquid, the change will occurs in melting point just on the solid material that convert to liquid material but if it solid material it needs to convert to another phase to be solid. The graphite is the most material that influenced by melting point in atmosphere, ti melted in 3948K, the mercury started melting at 234.32K. The melting point is used to find out the purity of organic material that the metal composed from.

Type of furnaces:

- Crucible furnace
- Dough furnace
- Turner furnace
- Kiln rotating furnace
- Induction Furnace
- Electric arc furnace

Crucible furnaces: Is the furnace that buried more than half underground, and the borders are coated by Thermal Bricks that contain 42% of alumina, and put inside it crucible made of graphite or zirconium, the flame is distributed through bottom gap connected to blower to insure the distribution of the flame.

Dough furnace: is the oldest furnace, it consist of coated long cylinder (6-11) m, semi circular based on iron base with four legs and two bottom doors to entering the coke and two openings for remove the melting and slag.

Electric arc furnace: is a modern techniques to produce the solid iron, these techniques are depends on the electrical current with high voltage to melting the sponge iron that resulting from the direct reduction.

Types of metals which are melting in the factory:

Iron: is a solid metal with brown or brownish red, extracted from rocks in nature, it influences of humidity then it oxidation so it must be coated to isolate it from weather condition. Melting the iron needs high temperatures to convert it from solid phase to liquid phase using catalysts to easy usage such as ships, rail ways and bridges industries.

Aluminum: is the most metal spread in the earth crust, it's a light metal and resists corrosion also it is easy to react with another material, it introduces in different industries. The bauxite ore is the major source of the aluminum, its good electric and thermal conductivity. It is used in planes, cars and trains industries. The aluminum is melting in two stages:

- Bauxite melting
- Alumina melting

Copper: is the most metal that the human uses in life, it introduced in alloys to produce a good and new properties of alloys. It has a good flexibility and good electric and thermal conductivity, also it is used in electric wires and decorations.

The catalyst:

- Fuels: is an organic material contains carbon and hydrogen are flammable, there are two types of fuels are used in furnaces: coke and natural gas.
- Catalyst: the most important catalyst is the limestone.

Project Aims

- Exploitation of the scrap containing different metals such as (Al, Cu, Fe) that spread in Mafraq region.
- Generate jobs for local community
- Produce industrial pieces with high quality
- Increase exporting of the Jordanian minerals

Suggested Service

The objective of the project is produce and forming pure iron, aluminum and copper and it will be melting using furnaces and moulds according to desired product such as pit of hummers, equipment of loaders, engineering and cars shapes.

Targeted Categories

The project is targeted some categories such as:

- Phosphate mines company
- Caterpillar company
- Quarry and mines
- water Authority
- National electric company
- Cars spare parts shops
- Accsesorries shops

Market Analysis

Based on the technical experience and knowledge of mould and alloys, the annual production of the working companies is about 6000 ton/year with different products (iron, copper, aluminum).

Compititors Analysis

The most competitors companies:

- Saba' company: producing hummer pit
- Mahmoud Alzoubi Company producing: loader equipments
- Retaaj Company: sewer cover products
- The techniqal company: specialized in copper and aluminum casting

Market Share

Based on the annual production of the molds company a factory will be established for melt and casting the metals with market share of about 10% and annual production 600 ton/year, hours/ day to provide the local markets of desired molds and it has a chance to produce move developed products which will increase the estimated revenues.

Prices Analysis

Table 3 Expected prices during years (2018-2027)

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Price of 1 ton of sewage coverings	600	606	612	618	624	631	637	643	650	656
Price of 1 ton of spare parts	3000	3030	3060	3091	3122	3153	3185	3216	3249	3281
Price of 1 ton of aluminium auto accessories	6000	6060	6121	6182	6244	6306	6369	6433	6497	6562
Price of 1 ton of accessories metal and carpenters of brass	7000	7070	7141	7212	7284	7357	7431	7505	7580	7656

Expected Revenues

Table4 Projected annual revenue for 10 years.

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
sewage Coverings revenues	118800	119988	121188	122400	123624	124860	126109	127370	128643	129930
spare parts revenues	306000	309060	312151	315272	318425	321609	324825	328073	331354	334668
aluminium auto accessories revenues	900000	909000	918090	927271	936544	945909	955368	964922	974571	984317
accessories metal and carpenters of brass revenues	1050000	1060500	1071105	1081816	1092634	1103561	1114596	1125742	1137000	1148370
Total of revenues	2374800	2398548	2422533	2446759	2471226	2495939	2520898	2546107	2571568	2597284

3. Technical analysis

The Human Resources Requirements

Table5 the human resources requirments

Position	Number
Project manager	1
Cast technican	2
technician maintenance	1
Accountant	1
Receptionist	1
Labour	2
Driver	2
Guard	2
Cleaning labour	1
Total	13

Job description of human resources:

Project manager: proposed the general strategy, prepare the plans and development of the project, supervise the staff, developing the procedures of safety and occupational health.

Receptionist: receiving the calls and customers.

Cast technican: carry out the casting process

Accountant: carry out the financial chek, audit, preparing the annual financial reports.

Guard: keep the security of the factory

Cleaning labour: keep the site and offices in good and cleaning shape.

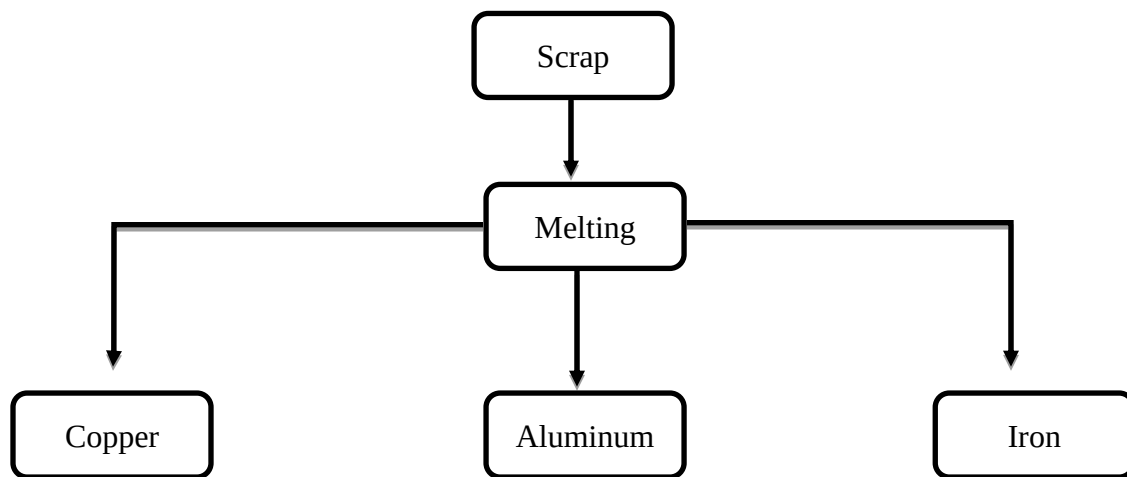
Location Analysis

The factory will be established at Mafraq governorate closer to the infrastructure service such as electricity, water and paving roads.

Description the Production Process

The metals are melting in special furnaces for each metal using catalyst then the casting and forming process will be carried out and as shown in the flowchart below:

Figure (3) flow sheat of process



Technical Requirements Analysis

1.1.1 Land and construction works

Purchasing the land in Mafraq governorate with an area of 10 dunoms the price of one dunom is estimated at 30,000 JOD's so the cost of land is 300,000 JOD's, the table 6 clarify shown the construction works required for the project:

Table6 the construction works

Construction works	Area m ²
Building works	200
Hunger building	1,000
Paving road	500
Another works	-



Equipments and Machines

Table (7) shows the equipments and machines required:

Table7 the equipments and machines

Equipments and machines	No.
Furnaces to melt the iron 2ton	1
Furnace to melt the copper 1ton	1
Furnace to melt the aliminum	1
Moulds casting	150
large crucible 50kg	20
medium crucible 25kg	30
small crucible 5kg	200
Simple furnace	1

Furnitures

The project needs integrated furniture, as administrative offices and workers rooms

Vehicles and Transportations

The following table (8) shows the list of vehicles and transportation required for the project

Table 8 vehicles of projects

Description	No.
Pick-up	2



Requirments and Legal Procedures

The following figure shows the necessary procedures for exploration and mining rights license depending on body regulating energy and minerals sectors:



4. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 12.05% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table9 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table10 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table11 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1)) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table12 Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	60.00%
Utilities Expenses from Total Revenues	2.5%
Maintenance Expenses from Total Revenues	2.50%
Depreciation Expenses from Total Revenues	1.00%
Insurance Expenses from Total Assets Value	0.75%
Marketing Expenses from Total Revenues	0.25%

Table13 Income Tax Assumptions

Income Tax Assumptions	Value
(1) Average Income Tax in Jordan	5%
(2) Income Tax Deduction	0.00
Income Tax after Deduction	5%
(3) Compulsary Reserve Percentage	10%
Other Assumptions	Value
(1) Annual Monthly Salaries have been calculated after multiplying monthly salaries by:	16

Risk Premiun	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	9.48%
(3) Market Risk Premiun	10.12%
Income Tax Rate	5.00%
(4) Beta	1
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table14 Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premiun	%10.12
Income Tax Rate	%5
Bets	1.00
Equity	567,620
Loans Value	851,430
Loans	

Weighted Average Cost of Capital	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%9.01
Loans Value	851,430
Loans Percentage	%60
Equity	
Cost of Equity	%16.6
Equity Value	567,620
Equity Percentage	%40
Gross	
Project Value	1,419,050
Weighted Average Cost of Capital	%12.05

Capital Expenditures

The estimated cost of the project is 1,419,050 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table15 land work Capital Expenditure

Description	Cost (JOD per Squared)	Area	Gross Cost
Land required	30	10,000	300,000
Total			300,000

Table16 Construction Work Capital Expenditure

Description	Cost (JOD per Squared)	Area	Gross Cost
Construction Works	200	200	40,000
Hungers works	100	1,000	100,000
Paving	15	500	7,500
Other Work			14,750
Total			162,250

Table17 machinery and equipments Capital Expenditure

Description	No.	Cost(JOD per unit)	Gross cost
Furnaces to melt the iron 2ton	1	130,000	130,000
Furnace to melt the copper 1ton	1	80,000	80,000
Furnace to melt the aliminum	1	40,000	40,000
Moulds casting	150	10	1,500
large crucible 50kg	20	500	10,000
medium crucible 25kg	30	300	9,000
small crucible 5kg	200	50	10,000
Simple furnace	1	30,000	30,000
Total			310,500

Table18 Furniture and furnishings Capital Expenditure

Description	No.	Cost(JOD per unit)	Gross cost
furniture	-	-	10,000
Total			10,000

Table19 vehicles and trasnportations Capital Expenditure

Description	No.	Cost(JOD per unit)	Gross cost
Medium vehicle (pickup)	2	23,000	46,000

Table20 Pre Operating Expenses

Pre-operating expenses	Estimaited cost (JD)
Governmental fees	500
Transportation	750
Marketing	500
Training	750
Total	2,500

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table21 Working Capital

Working Capital	No. of	%	Estimated Cost
Operating expenses	3	%25	431,026
General and administrative expenses	3	%25	3,805
Marketing expenses	3	%25	1,484
Indirect Salaries	3	%25	22,480
Total			458,796

Table22 Expenses Summary

Capital expenditure	Cost (JD)	Contingency	Cost (JD)	%
Land	300,000	10%	330,000	%23.3
Construction works	162,250	10%	178,475	%12.6
Machinery & Equipment	310,500	10%	341,550	%24.1
Furniture	10,000	10%	11,000	%0.8
Vehicles	46,000	10%	50,600	%3.6
Pre-Operating Expenses	2,500	10%	2,750	%0.2
Working capital	458,796	10%	504,675	%35.6
Total (JD)	1,290,046		1,419,050	100%

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (1,419,050) JD, where 60% of the project will be financed through loans (851,430) JD, and the remaining (40%) through (Equity) with (567,620) JD. The following table summarizes the general structure of the required financing:

Table23 Sources of Funding

Funding Sources	Amount	Percentage
Equity	567,620	%40
Loans	851,430	%60
Total	1,419,050	%100
Use of Fund:	Value	Ratio
Capital Expenditures	911,625	%64.24
Pre-Operating Expenses	2,750	%0.19
Working Capital	504,675	%35.56
Total	1,419,050	%100.00

Operating Expenses

The following table summarizes the expected operating expenses for the proposed project (2018-2027):

Table 24 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials and Packaging from Revenues	1,424,880	1,439,129	1,453,520	1,468,055	1,482,736	1,497,563	1,512,539	1,527,664	1,542,941	1,558,370
Utilities from Revenues	59,370	59,964	60,563	61,169	61,781	62,398	63,022	63,653	64,289	64,932
Maintenance from Revenues	59,370	59,964	60,563	61,169	61,781	62,398	63,022	63,653	64,289	64,932
Disposables from Revenues	23,748	23,985	24,225	24,468	24,712	24,959	25,209	25,461	25,716	25,973
Others	156,737	158,304	159,887	161,486	163,101	164,732	166,379	168,043	169,723	171,421
Total	1,724,105	1,741,346	1,758,759	1,776,347	1,794,110	1,812,051	1,830,172	1,848,474	1,866,958	1,885,628

General and Administrative Expenses

The following table summarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 25 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	1,200	1,224	1,248	1,273	1,299	1,325	1,351	1,378	1,406	1,434
Hospitality	600	612	624	637	649	662	676	689	703	717
Stationary and Printing	600	612	624	637	649	662	676	689	703	717
Transportation	3,600	3,672	3,745	3,820	3,897	3,975	4,054	4,135	4,218	4,302
Training	1,000	1,020	1,040	1,061	1,082	1,104	1,126	1,149	1,172	1,195
Insurance	6,837	6,974	7,113	7,256	7,401	7,549	7,700	7,854	8,011	8,171
Miscellaneous	1,384	1,411	1,440	1,468	1,498	1,528	1,558	1,589	1,621	1,654
Total	15,221	15,525	15,836	16,153	16,476	16,805	17,141	17,484	17,834	18,190

Marketing Expenses

Table 26 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	5,937	5,996	6,056	6,117	6,178	6,240	6,302	6,365	6,429	6,493

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table27 The projected staff number

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Project manager	1	1	1	1	1	1	1	1	1	1
Cast technical	2	2	2	2	2	2	2	2	2	2
Technical maintenance	1	1	1	1	1	1	1	1	1	1
Accounter	2	2	2	2	2	2	2	2	2	2
Receptions	2	2	2	2	2	2	2	2	2	2
Labor	2	2	2	2	2	2	2	2	2	2
Driver	1	1	1	1	1	1	1	1	1	1
Guard man	1	1	1	1	1	1	1	1	1	1
Cleaner man	1	1	1	1	1	1	1	1	1	1
Total	13	13	13	13	13	13	13	13	13	13

Table 28 Expected monthly salary and wages for the years (2018-2027)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Project manager	1,500	1,562	1,626	1,692	1,762	1,834	1,909	1,987	2,069	2,154
Cast technical	380	396	412	429	446	465	484	503	524	546
Accounter	400	416	433	451	470	489	509	530	552	574
Labor	350	364	379	395	411	428	445	464	483	502
Driver	380	396	412	429	446	465	484	503	524	546
Guard man	300	312	325	338	352	367	382	397	414	431
Cleaner man	250	260	271	282	294	306	318	331	345	359
Technical maintenance	350	364	379	395	411	428	445	464	483	502
Receptions	300	312	325	338	352	367	382	397	414	431

Table 29 Expected Annual Salaries and Wges (2027-2018)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Project manager	24,000	24,984	26,008	27,075	28,185	29,340	30,543	31,796	33,099	34,456
Cast technical	12,160	12,659	13,178	13,718	14,280	14,866	15,475	16,110	16,770	17,458
Accounter	6,400	6,662	6,936	7,220	7,516	7,824	8,145	8,479	8,826	9,188
Labor	11,200	11,659	12,137	12,635	13,153	13,692	14,254	14,838	15,446	16,080
Driver	12,160	12,659	13,178	13,718	14,280	14,866	15,475	16,110	16,770	17,458
Guard man	9,600	9,994	10,403	10,830	11,274	11,736	12,217	12,718	13,240	13,782
Cleaner man	4,000	4,164	4,335	4,512	4,697	4,890	5,091	5,299	5,517	5,743
Technical maintenance	5,600	5,830	6,069	6,317	6,576	6,846	7,127	7,419	7,723	8,040
Receptions	4,800	4,997	5,202	5,415	5,637	5,868	6,109	6,359	6,620	6,891
Project manager	89,920	93,607	97,445	101,440	105,599	109,928	114,435	119,127	124,012	129,096

Depreciations

The following tables illustrate cost of Capex and Depreciation rates:

Table30 Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	330,000	%0.0	%0.0
Construction Works	178,475	%5.0	%0.0
Machinaries	341,550	%10.0	%2.0
Furniture	11,000	%10.0	%5.0
Vehicles	50,600	%10.0	%0.0

Table31 Depreciations and Additions on Construction Works

Capex, Annual Additions and Depreciations	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	911,625	919,006	926,551	934,264	942,150	950,213	958,457	966,887	975,507	984,323
Total Depreciation Expenses	49,239	49,977	50,731	51,503	52,291	53,098	53,922	54,765	55,627	56,509
Total Accumulated Depreciation	49,239	99,216	149,947	201,450	253,741	306,838	360,760	415,525	471,152	527,661
Total Additions	0	7,381	7,545	7,713	7,886	8,063	8,244	8,430	8,621	8,816
Total Net Book Values	862,386	819,790	776,604	732,815	688,409	643,374	597,696	551,361	504,355	456,663

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table 32 Loan Details

Loan Value	851,430			
Annual Interest Rate	9.48%			
Loans Period	5			
Loan Starts at year:	2018			

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				851,430
2019	0	80,716	-80,716	932,146
2020	0	88,367	-88,367	1,020,513
2021	0	96,745	-96,745	1,117,258
2022	0	105,916	-105,916	1,223,174
2023	0	115,957	-115,957	1,339,131
2024	0	126,950	-126,950	1,466,080
2025	0	138,984	-138,984	1,605,065
2026	0	152,160	-152,160	1,757,225
2027	0	166,585	-166,585	1,923,810
2028	0	182,377	-182,377	2,106,187

Income Statment

Table 33 Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	2,374,800	2,398,548	2,422,533	2,446,759	2,471,226	2,495,939	2,520,898	2,546,107	2,571,568	2,597,284
Gross Operating Revenues	2,374,800	2,398,548	2,422,533	2,446,759	2,471,226	2,495,939	2,520,898	2,546,107	2,571,568	2,597,284
Operating Expenses	-1,724,105	-1,741,346	-1,758,759	-1,776,347	-1,794,110	-1,812,051	-1,830,172	-1,848,474	-1,866,958	-1,885,628
Gross Operating Profit(	650,695	657,202	663,774	670,412	677,116	683,887	690,726	697,633	704,610	711,656
<i>Gross Profit Percentage</i>	<i>%27</i>	<i>%27</i>	<i>%27</i>	<i>%27</i>	<i>%27</i>	<i>%27</i>	<i>%27</i>	<i>%27</i>	<i>%27</i>	<i>%27</i>
Salaries and Benefits (Indirect Staff)	-89,920	-93,607	-97,445	-101,440	-105,599	-109,928	-114,435	-119,127	-124,012	-129,096
General and Administraive Expenses	-15,221	-15,525	-15,836	-16,153	-16,476	-16,805	-17,141	-17,484	-17,834	-18,190
Markeing Expenses	-5,937	-5,996	-6,056	-6,117	-6,178	-6,240	-6,302	-6,365	-6,429	-6,493
Pre Operating Expenses	-2,750									
Gross Indirect Expenses	-113,828	-115,128	-119,337	-123,709	-128,253	-132,973	-137,879	-142,977	-148,274	-153,780
Income before Interest, Depreciation and Tax	536,867	542,074	544,437	546,703	548,864	550,914	552,847	554,657	556,335	557,876
Fixed Assets Depreciations	-49,239	-49,977	-50,731	-51,503	-52,291	-53,098	-53,922	-54,765	-55,627	-56,509
Income before Tax and Interests	487,629	492,097	493,706	495,200	496,572	497,816	498,925	499,892	500,708	501,368

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Bank Interests	-80,716	-88,367	-96,745	-105,916	-115,957	-126,950	-138,984	-152,160	-166,585	-182,377
Income Before Tax	406,913	403,729	396,961	389,284	380,615	370,867	359,941	347,732	334,124	318,990
Income Tax	-20,346	-20,186	-19,848	-19,464	-19,031	-18,543	-17,997	-17,387	-16,706	-15,950
Net Profit	386,567	383,543	377,113	369,820	361,585	352,323	341,944	330,345	317,417	303,041
<i>Net Profit Percentage</i>	<i>%16</i>	<i>%16</i>	<i>%16</i>	<i>%15</i>	<i>%15</i>	<i>%14</i>	<i>%14</i>	<i>%13</i>	<i>%12</i>	<i>%12</i>
Compulsory Reserves	-38,657	-38,354	-37,711	-36,982	-36,158	-35,232	-34,194	-33,035	-31,742	-30,304
Retained Earnings	347,911	693,099	1,032,501	1,365,339	1,690,765	2,007,856	2,315,606	2,612,916	2,898,592	3,171,329

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table34 Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities										
Net Profit	386,567	383,543	377,113	369,820	361,585	352,323	341,944	330,345	317,417	303,041
Bank Interests	80,716	88,367	96,745	105,916	115,957	126,950	138,984	152,160	166,585	182,377
Depreciation	49,239	49,977	50,731	51,503	52,291	53,098	53,922	54,765	55,627	56,509
Total Operating Cashflows before Additions to Working Capital	516,522	521,887	524,589	527,238	529,833	532,370	534,850	537,270	539,629	541,927
Inventory (Increase/Decrease)	-118,740	-1,187	-1,199	-1,211	-1,223	-1,236	-1,248	-1,260	-1,273	-1,286
Accounts Receivables (Increase/Decrease)	49,871	499	504	509	514	519	524	529	535	540
Working Capital (Increase/Decrease)	-68,869	-689	-696	-703	-710	-717	-724	-731	-738	-746
Net Cashflows from Operating Activities	447,652	521,199	523,894	526,536	529,123	531,654	534,126	536,539	538,891	541,181
Cashflows from Investments Activities										
Fixed Assets (Procurement)	-911,625	-7,381	-7,545	-7,713	-7,886	-8,063	-8,244	-8,430	-8,621	-8,816
Net Cashflows from Investments	-911,625	-7,381	-7,545	-7,713	-7,886	-8,063	-8,244	-8,430	-8,621	-8,816
Cashflows from Financing Activities										
Capital	567,620									

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Return the loan	80,716	88,367	96,745	105,916	115,957	126,950	138,984	152,160	166,585	182,377
Bank Interest Rate	-80,716	-88,367	-96,745	-105,916	-115,957	-126,950	-138,984	-152,160	-166,585	-182,377
Loans	851,430	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing Activities	1,419,050	0	0	0	0	0	0	0	0	0
Net (Increase/Decrease) in Cash	955,078	513,818	516,349	518,823	521,237	523,591	525,882	528,109	530,270	532,365
Cashflows at the Beginning of Period	0	955,078	1,468,895	1,985,244	2,504,066	3,025,304	3,548,895	4,074,777	4,602,887	5,133,157
Cashflows at the End of Period	955,078	1,468,895	1,985,244	2,504,066	3,025,304	3,548,895	4,074,777	4,602,887	5,133,157	5,665,522

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 35 Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current Assets										
Cash	955,078	1,468,895	1,985,244	2,504,066	3,025,304	3,548,895	4,074,777	4,602,887	5,133,157	5,665,522
Inventory	118,740	119,927	121,127	122,338	123,561	124,797	126,045	127,305	128,578	129,864
Total current Assets	1,073,818	1,588,823	2,106,371	2,626,404	3,148,865	3,673,692	4,200,822	4,730,192	5,261,735	5,795,386
Non Current Assets										
Fixed Assets (net)	862,386	819,790	776,604	732,815	688,409	643,374	597,696	551,361	504,355	456,663
Total Non Current Assets	862,386	819,790	776,604	732,815	688,409	643,374	597,696	551,361	504,355	456,663
Total Assets	1,936,204	2,408,613	2,882,975	3,359,219	3,837,275	4,317,066	4,798,519	5,281,553	5,766,090	6,252,048
Liabilities										
Current liabilities										
Payables	49,871	50,370	50,873	51,382	51,896	52,415	52,939	53,468	54,003	54,543
Remaining amount of Loan	88,367-	96,745-	105,916-	115,957-	126,950-	138,984-	152,160-	166,585-	182,377-	0
Total current Liabilities	38,497-	46,375-	55,043-	64,575-	75,054-	86,570-	99,221-	113,117-	128,374-	54,543
Non Current Liabilities										

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Long Terms Loans	1,020,513	1,117,258	1,223,174	1,339,131	1,466,080	1,605,065	1,757,225	1,923,810	2,106,187	2,106,187
Total Long Term Liabilities	1,020,513	1,117,258	1,223,174	1,339,131	1,466,080	1,605,065	1,757,225	1,923,810	2,106,187	2,106,187
Total Liabilities	982,017	1,070,883	1,168,131	1,274,556	1,391,026	1,518,495	1,658,004	1,810,693	1,977,813	2,160,730
Owners Equity										
Shareholders Contributions	567,620	567,620	567,620	567,620	567,620	567,620	567,620	567,620	567,620	567,620
Statutory Reseve	38,657	77,011	114,722	151,704	187,863	223,095	257,290	290,324	322,066	352,370
Retained Profits	347,911	693,099	1,032,501	1,365,339	1,690,765	2,007,856	2,315,606	2,612,916	2,898,592	3,171,329
Total Equity	954,187	1,337,730	1,714,844	2,084,663	2,446,248	2,798,571	3,140,515	3,470,860	3,788,278	4,091,318
Total liabilities and equity	1,936,204	2,408,613	2,882,975	3,359,219	3,837,275	4,317,066	4,798,519	5,281,553	5,766,090	6,252,048

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 36 Free Net Cash flows Table

Net Cashflows	Pre- Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final value
Net Free Cashflows	-1,419,050	450,402	513,818	516,349	518,823	521,237	523,591	525,882	528,109	530,270	532,365	6,876,388
Discount Factor	1.00	0.71	0.63	0.57	0.51	0.45	0.40	0.36	0.32	0.29	0.26	0.26
Net Present Value for Cash Flows	-1,419,050	320,145	325,939	292,316	262,126	235,022	210,692	188,854	169,256	151,670	135,892	1,755,269

Table below (37) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table37 Payback Period

Payback Period	Pre- Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	-1,419,050	450,402	513,818	516,349	518,823	521,237	523,591	525,882	528,109	530,270	532,365
Project Value Returned	1,419,050	968,648	454,830	-61,518	-580,341	-1,101,578	-1,625,170	-2,151,052	-2,679,161	-3,209,431	-3,741,796
Payback Period (Year)	2	1	1	0	0	0	0	0	0	0	0

Table 38 Financial Analysis Results

Indicator	Value
Weighted Average Cost of Capital (WACC)	%12.05
Net Present Value for Cashflows	2,628,130
Payback Period	2
Internal Rate of Return	%38.92

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Table 39 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%38.92	2	12.05%
Revenues declined by 10%	%35.91	3	12.05%
Operating Expenses Increased by 10%	%27.63	4	12.05%

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 2,628,130 Jordan Dinars considering that the project provides 13 Job Opportunities for the governorate residents.